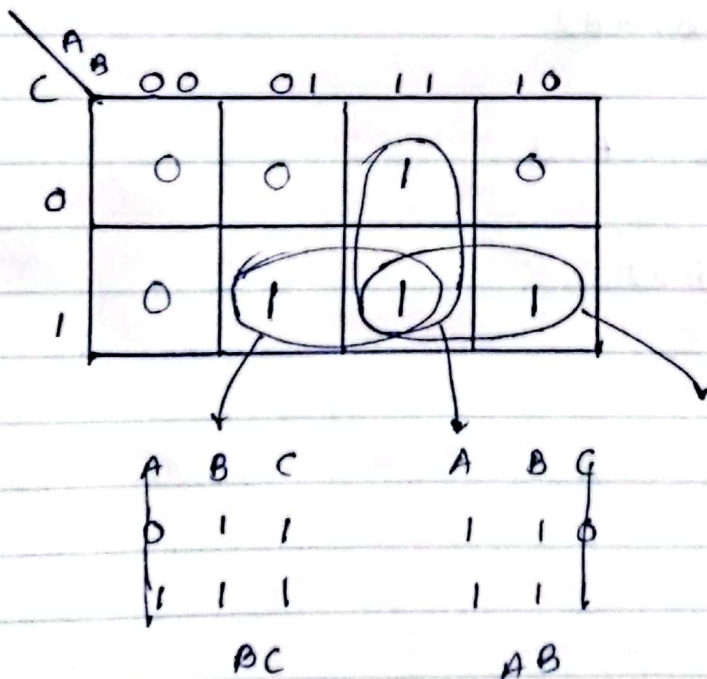


Full Adder

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum

$$\text{Carry} = \bar{A}BC + A\bar{B}C + ABC\bar{C} + ABC$$



$$\text{Carry} = \bar{A}BC + A\bar{B}C + ABC\bar{C} + ABC$$

Carry =

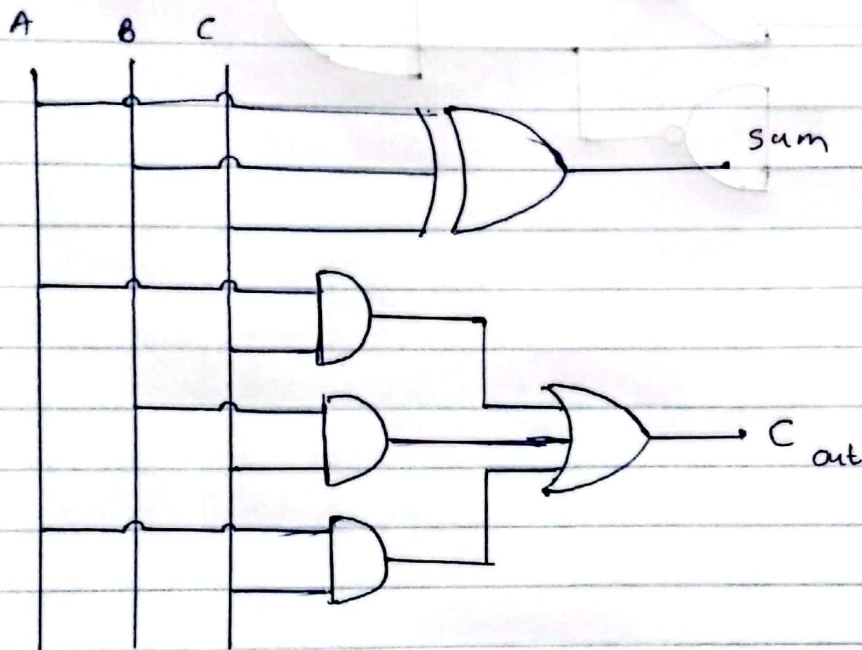
$$\begin{aligned} \text{Sum} &= \bar{A}\bar{B} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\ &= \bar{A}(\bar{B}\bar{C} + B\bar{C}) + A(\bar{B}\bar{C} + BC) \\ &= \bar{A}(B \oplus C) + A(\overline{B \oplus C}) \end{aligned}$$

$$\text{if } B \oplus C = D$$

$$\bar{A}D + A\bar{D}$$

$$A \oplus D$$

$$\therefore \underline{\underline{A \oplus B \oplus C}}$$



★ Full Adder using NAND gates

